Progressive decrease in effective dimensionality of visual cortex regions

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1 Introduction

Creating artificial neural networks (ANNs) that are aligned to observed brain activity could provide a testbed to perform neuroscience experiments that are infeasible in living subjects.

Though strategies exist to quantify brain-ANN similarity [1], it is unclear how the underlying structure of data influences these metrics. Differences in the effective dimensionality of neural responses or model weights, for example, may influence the current approach to similarity scoring.

Characterizing the dimensionality of brain regions is contentious, with investigators drawing contrasting conclusions about whether the visual cortex is better characterized as high- or low-dimensional [2, 3]. The source of divergence remains unclear, and possible causes include differences in neural recording, task complexity, and choice of statistical method. Furthermore, the effect of dimensionality on model similarity is also controversial, with investigators disagreeing about whether high effective dimensionality drives higher similarity scores [4, 5].

Within this project, I will perform a comprehensive investigation of data collected in the macaque visual cortex [6]. Because researchers presented upwards of 1000 unique stimuli and recorded activity from different visual regions from the same subjects, I anticipate that this data set will contain sufficient detail to address existing questions about neural dimensionality.

Table 1: Description of the macaque monkey subjects and relevant features of the recording data. For monkeys with more than one recording over time, the youngest and oldest age recording, in weeks, is reported as Age 1 and Age 2, respectively.

Monkey ID	Sex	Age 1	Age 2	Region(s)
M1	F	17	95	V1-V2 border, V4
M2	F	16	95	V2, V4
M3	M	16	65	FC, IT
M4	F	27	53	NA
M5	M	17	64	NA
M6	M	184	194	V1, IT
M7	M	184	194	IT
M8	M	NA	NA	V4

2 Data

Lee et al. trained 8 macaque monkeys to fixate on a screen and presented the animals with a passive fixation tasks. The experimenters placed 96-electrode "Utah" recording arrays in 8 macaque monkeys, producing data from visual regions V1, V2, V4, IT, and FC (foveal confluence). For monkeys M2, M3, and M6, a second electrode produced parallel recordings in two visual regions, seen in Table 1. Lee et al. published the binned spiking data, summed for bins of 10 ms, but omitted the raw spike timing data.

The monkeys viewed synthetic textures samples with natural structure ("naturalistic") or without structure ("noise") [7]. A subset of monkeys also performed a 4-choice task to discriminate naturalistic and noise samples, though this behavioral data will likely not be relevant for the project. Monkeys viewed images in blocks of 8, with images shown for 100 ms and separated by 100 ms of blank stimulus. Generally, later visual regions show greater relative divergence between noise and naturalistic images, seen in Figure 1.

3 Sources

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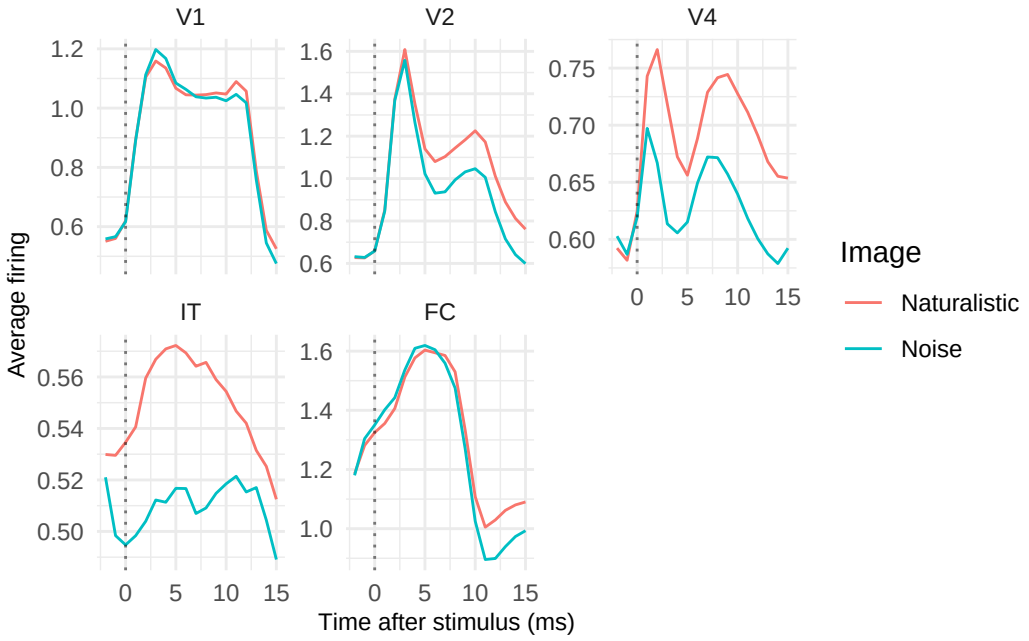

Figure 1: Average firing rate over images with naturalistic (red) or noise (blue) structure, separated by region in the visual cortex. In general, later visual regions (V4, IT) tend to show more divergence than early visual regions.

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